Trajectory following using AL-iLQR method and obstacle avoidance using constraint

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Abstract

In this project, obstacle avoidance for UAV (Unmanned Aerial Vehicles) is explored using the iLQR (iterative Linear Quadratic Regulator method). As, obstacle avoidance is very important in the UAV field because it is undeniable. Since, there are many algorithm which can be used for obstacle avoidance like MPC (Model Predictive Control), APF (Artificial Potential Field) and others. MPC is powerful and can be used with non-linear system efficiently but here we will be using AL-iLQR (Augmented Lagrangian iLQR) method. This algorithm help us update the state space while considering the constraint with time.

Keywords: AL-iLQR, UAV, Obstacle avoidance,

1. Introduction

Obstacle avoidance in UAVs is a challenge in autonomous flight control, especially in dynamic or constrained environments. To ensure safe navigation and optimizing flight performance at the same time, the best control algorithm for real-time decision making in the presence of obstacles is required. One method is the Iterative Linear Quadratic Regulator (iLQR), particularly when combined with constraint-handling algorithm like the Augmented Lagrangian (AL), Kawarabayashi (2023). These techniques offer a balance between computational efficiency and the ability to handle non-linear dynamics and constraints. LQR is widely used for UAV trajectory tracking but lacks support for constraints and nonlinear dynamics, Kawarabayashi (2023). iLQR with Augmented Lagrangian (AL-iLQR) allows real-time trajectory optimization with obstacle avoidance and constraint handling for quadrotors. Constrained iLQR improves collision risk assessment using geometric methods like collision polygons, effective in autonomous driving and applicable to UAVs. CILQG extends iLQR to belief-space planning under uncertainty, enhancing safety in noisy or dynamic environments, Chen et al. (2021). MPC with LQR-based tuning offers constraint handling and optimal control for UAV collision avoidance with real-time feasibility, Aoki et al. (2021). DDP and MPPI are effective for complex UAV dynamics and urban air mobility, showing real-time path planning potential, Houghton et al. (2022).

2. Model and Problem Formulation

2.1. Quadrotor Dynamics

Quadrotor is a highly nonlinear system with 6 DOF (degree of freedom). In this study, we have linearized the quadrotor dynamics for our simplicity. We have only considered 2D motion of the quadrotor in XY plane while ignoring the altitude effect and yaw angle. Quadrotor is asummed to be symmetrical, rotor gyroscopic effect is neglected, and it flies at higher height which remove the ground effect, Martins et al. (2019).

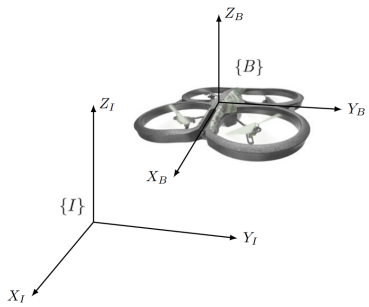

Figure 1: New Astronomy Journal cover.

The equation of motion for the quadrotor is given below (according to

$$m\ddot{p} = -mg\vec{a}_3 + \mathbb{R}_B\vec{B} \tag{1}$$

$$\mathbb{I}\dot{\omega} = -\omega \times \mathbb{I}\omega + \tau \tag{2}$$

Where $\mathbb{I}$ is the quadrotor inertia matrix, $\boldsymbol{\omega}$ is the angular velocity, $\boldsymbol{\tau}$ is the torque produced by the rotors, $\mathbb{R}$ is the rotation matrix in the body frame and $\vec{a}_3$ represents the direction of the gravitational force.

2.2. Linearization

As we discussed, the dynamics is simplified so for our futhur study it is linearized about the hover condition. Now, let $P = [x, y]'$ be the position of the quadrotor in 2D, $P = [u, v]'$ be the velocity and assume that change in angle is very small so all the *cosine* and *sine* be equal to 1 and θ respectively, Martins et al. (2019).

In paper Martins et al. (2019), the system dynamics is studied by creating subsystem for different dimension like for linear motion and angular motion. For our study, we will only study

the linear motion dynamics. So the state space equation for 2D linear motion is given below:

$$\cos\theta \approx 1, \quad \sin\theta \approx \theta \quad \text{for small } \theta \tag{3}$$

$$\ddot{\theta} \approx \frac{\tau_\theta}{I_y} \tag{4}$$

$$x_\theta = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}, \quad u_\theta = \tau_\theta \tag{5}$$

$$\dot{x}_\theta = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x_\theta + \begin{bmatrix} 0 \\ 1/I_y \end{bmatrix} u_\theta \tag{6}$$

$$\ddot{\phi} \approx \frac{\tau_\phi}{I_x} \tag{7}$$

$$x_\phi = \begin{bmatrix} \phi \\ \dot{\phi} \end{bmatrix}, \quad u_\phi = \tau_\phi \tag{8}$$

$$\dot{x}_\phi = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x_\phi + \begin{bmatrix} 0 \\ \frac{1}{I_x} \end{bmatrix} u_\phi \tag{9}$$

The acceleration of the quadrotor in the body frame is given by the equation (10)

$$\vec{a}_B = \frac{\vec{F}}{m} - \mathbb{R}_B \vec{g}_{a_3} - \omega \times \vec{v}_B \tag{10}$$

This equation is linearized using above assumption and thus we get

$$a_x = \theta_g, \quad a_y = -\phi_g \tag{11}$$

After combining both two subsystem to form one system for 2D motion analysis we get

$$\begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{u} \\ \dot{v} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ u \\ v \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ g & 0 \\ 0 & g \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} \tag{12}$$

Since, we got the linearized system dynamics in state-space form. Now, it is discretized using the Euler's approximation method

$$A_s = I + AT = \begin{pmatrix} 1 & T \\ 0 & 1 \end{pmatrix} \tag{13}$$

$$B_s = BT \tag{14}$$

So, the matrix A and B we got after discretization is given below:

$$A = \begin{pmatrix} 1.0 & 0.0 & 0.01 & 0.0 \\ 0.0 & 1.0 & 0.0 & 0.01 \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 1.0 \end{pmatrix}$$

(15)

$$B = \begin{pmatrix} 0.0 & 0.0 \\ 0.0 & 0.0 \\ 0.01 & 0.0 \\ 0.0 & 0.01 \end{pmatrix} \tag{16}$$

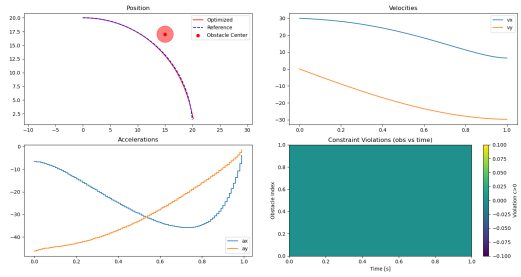

Figure 2: Trajectory following without obstacles

2.3. ILQR method

Since, the system dynamics is discretized we are ready to use that dynamics in iLQR method for trajectory following. We are going to use the algorithm as explained in the paper, Kawarabayashi (2023).

The system equation is

$$\dot{x} = f(x, u) \tag{17}$$

This can be discretized using time as

$$x_{k+1} = f(x_k, u_k, \Delta t) \tag{18}$$

The equation for the cost function is given by

$$V_k(x) = \min_{u_k} \left(\frac{1}{2} x_k^T Q_k x_k + \frac{1}{2} u_k^T R_k u_k + V_{k+1}(A_k x_k + B_k u_k) \right) \tag{19}$$

$$P_k = Q_k + A_k^T P_{k+1} A_k - A_k^T P_{k+1} B_k (R_k + B_k^T P_{k+1} B_k)^{-1} B_k^T P_{k+1} A_k \tag{20}$$

$$u_k^* = -(R_k + B_k^T P_{k+1} B_k)^{-1} B_k^T P_{k+1} A_k x_k \tag{21}$$

2.4. Constraint

The iLQR method above help us derive the trajectory for the quadrotor but it will not avoid the obstacles, where using constraint which is allowed in AL-iLQR method help us detect the obstacle and find the optimal trajectory using constraint by find the collision free path. The equation for the AL-iLQR method is taken from the paper, Kawarabayashi (2023).

$$c_{i,k} = (r_{\text{veh}} + r_i)^2 - \|\mathbf{p}_{i,k} - \mathbf{p}_k\|^2 \leq 0 \tag{22}$$

The above equation, Kawarabayashi (2023), help us avoid the collision. If it is negative then it is collision free otherwise there is collision.

3. Results

4. Discussion

The AL-iLQR method was applied to solve the trajectory optimization problem for a quadrotor, which translational motion while respecting state and input constraints. Compared to classical LQR and the AL-iLQR approach successfully generates collision-free paths even in dynamic environments with multiple obstacles. iLQR improved performance but still led to

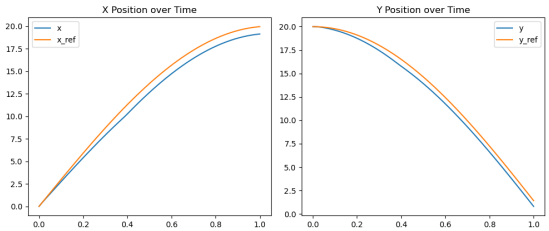

Figure 3: Position of quadrotor with time (x, t), (y, t)

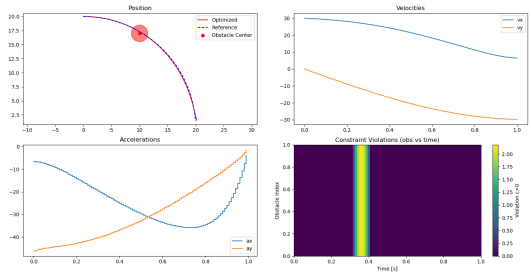

Figure 4: Not able to avoid collision due to higher penalty in Q and R

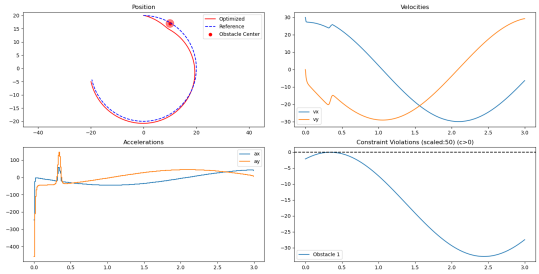

Figure 5: Position of quadrotor with time R = [0.001, 0.01], Q = [20, 20, 30, 40]

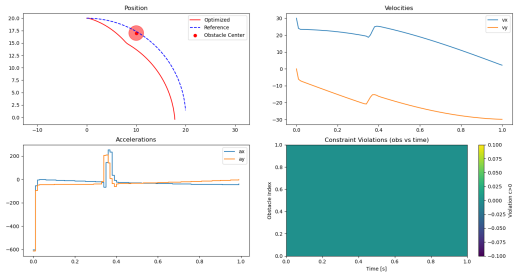

Figure 6: When the penalty is higher in R = [1, 1]

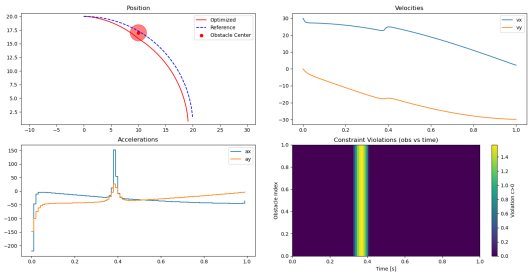

Figure 7: Constraint Violation is done but without collision

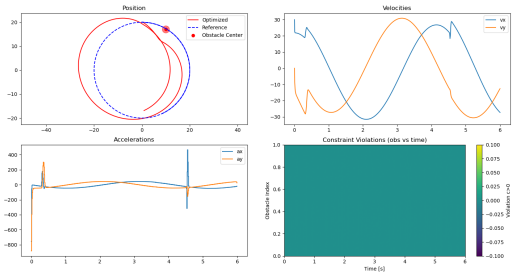

Figure 8: For longer duration

collisions when constraints were not enforced. AL-iLQR, by contrast, maintained constraint satisfaction and prevented collisions.

Despite its effectiveness, AL-iLQR exhibited higher computational demands, especially in early iterations or when the initial trajectory did not satisfy constraints. However, when the initial guess already adhered to constraints, the computation time was comparable to iLQR. The algorithm used penalty updates to improve constraint accuracy, and convergence was achieved using an iterative penalty increase approach. The quadrotor was able to reach target positions and avoid obstacles with predicted trajectories.

5. Summary and conclusions

The study demonstrates that AL-iLQR is a viable solution for constrained trajectory optimization in quadrotors. It allows for longer terminal positions and robust collision avoidance by incorporating constraints directly into the optimization. Although it takes several seconds to compute optimal inputs, making real-time applications challenging, the method is promising when initialized with feasible guesses. Future improvements should aim at including the dynamic obstacles and using MPC with other constraints and kalman filter.

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